AI-Accelerated Reconstruction for ATLAS TileCal

GSoC 2026 Evaluation Test Report

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Abstract

This report details a linear algorithmic approach to energy reconstruction for the ATLAS Tile Calorimeter (TileCal) under High-Luminosity LHC (HL-LHC) conditions. We explore and contrast a physics-driven Optimal Filtering (OF) style estimator with statistical data-driven solutions via Ridge Regression and Weighted Least Squares (WLS). A hyperparameter scan reveals the optimum regularization strength to be $\lambda = 0$, indicating that the final proposed model effectively reduces to unregularized Ordinary Least Squares (OLS). This model achieves a relative energy resolution of approximately 45% at the 10 MeV threshold and 5% at higher energies (> 50 MeV). We empirically observe that it maintains largely unbiased physics results across the dynamic range. These implementations are selected as the primary estimator, serving as a baseline for future development of low-latency Machine Learning models intended for FPGA deployment.

1 Introduction

The ATLAS Tile Calorimeter (TileCal) is an essential component of the ATLAS experiment at CERN, functioning as the central hadronic sampling calorimeter. During the upcoming High-Luminosity LHC (HL-LHC) era, the average number of inelastic proton-proton collisions per bunch crossing (pile-up, μ) will drastically increase. This environment demands highly precise and low-latency online energy and time reconstruction of detector signals sampled at 40 MHz (25 ns intervals).

The evaluation task requires reconstructing a scalar low-gain energy estimate, `ene_lo`, from a 7-sample window of digital ADC readouts (`sample_lo`) centered around the bunch crossing. Crucially, the algorithm must be strictly linear, mirroring the fixed-weight constraints of current real-time data-processing pipelines, yet designed to optimally reject pedestal drift, out-of-time pile-up, and electronic noise.

2 Related Work

TileCal currently relies heavily on the Optimal Filtering (OF) algorithm. The OF2 method analytically derives weights that produce unbiased amplitude and time estimates by projecting out the reference pulse shape alongside its derivative, while minimizing the variance caused by electronics noise and pile-up (captured via a noise covariance matrix) and enforcing immunity to constant pedestal offsets. As pile-up increases, more sophisticated analytical linear filters such as the Wiener filter have been studied for Run-3.

Recently, there has been a significant push towards replacing linear filters with Machine Learning (ML) approaches, including Multilayer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs). These ML models demonstrate superior resolution under extreme HL-LHC pile-up and can be heavily quantized for ultra-low latency execution on Field Programmable Gate Arrays (FPGAs). Our linear baseline provides the foundational mathematical architecture that these neural networks build upon and attempt to surpass.

3 Dataset and Problem Definition

The provided dataset mimics a simplified TileCal scenario:

- **Given:** A 7-dimensional feature vector $x_n \in \mathbb{R}^7$, representing time samples evaluated every 25 ns around the nominal bunch crossing.
- **Target:** A physical energy amplitude scalar $y_n = \texttt{ene_lo}[n]$.
- **Constraint:** The estimator must be defined as $\hat{y}_n = w^\top x_n + b$, where w and b are fixed learned parameters.

While the evaluation brief describes reconstructing the target from an 8-sample window, the actual provided PyTorch datasets structure the digital values into 7-sample sequences. Standard TileCal optimal filtering typically utilizes 7 consecutive samples. Our implementation strictly adheres to the 7-sample structure natively present in the provided shards.

Given that the brief requests the metric $r = (\hat{y} - y)/y$, evaluating performance near $y \approx 0$ is numerically unstable and physically uninformative to purely signal-oriented resolution. We introduce a signal threshold (e.g., 10 MeV) to evaluate relative error strictly on events representing genuine physics deposits rather than background pedestal fluctuations.

4 Linear Algorithm Design

To provide a rigorous solution, we investigated two distinct theoretical approaches:

4.1 Physics-Driven OF-Style Estimator

Following traditional TileCal Optimal Filtering 2 (OF2) methodology, we isolated a high-energy subset of the dataset to average a clean 7-sample **template pulse** g. We also calculated its discrete derivative g', and in low-energy regions, we calculated the 7×7 **noise covariance matrix** C.

Analytically, the optimal OF2 amplitude weights minimize variance $w^\top C w$ subject to three Lagrange constraints:

$$w^\top g = 1, \quad w^\top g' = 0, \quad \sum w_i = 0$$

This rigorous formulation ensures unbiased amplitude recovery, immunity to phase jitter, and exact cancellation of flat pedestal shifts caused by pile-up. Because the raw OF amplitude is obtained in template-normalized units, we apply a final linear calibration on the training set to map the estimate into the target energy units.

4.2 Regularized Linear Regression Scan, Final Selected Model: OLS

Alternatively, we formulated the reconstruction directly as a supervised regression task using Ordinary Least Squares with L_2 regularization. The objective is to minimize:

$$\min_{w,b} \sum_n \left(w^\top x_n + b - y_n \right)^2 + \lambda \|w\|^2$$

This data-driven approach effectively absorbs both the pulse shape *and* the complex, potentially non-Gaussian pile-up topology directly from the dataset, generalizing the analytic OF method.

Exploration of Weighted Least Squares (WLS): Because the evaluation metric is relative error, we experimented with WLS using weights $\propto 1/y^2$. While this mathematically optimizes the low-energy bins, it introduces an unsustainable systematic bias (over -60% under-estimation) at higher energies. Consequently, standard Ridge Regression was selected to ensure an operationally unbiased physics outcome across all energy scales.

5 Results and Discussion

The models were trained on the provided train shard, tuned on the provided validation shard, and evaluated on the provided test shard. The optimal penalty was found to be $\lambda = 0.0$, indicating that within the tested family of affine models on this dataset, additional L_2 regularization does not improve validation or test performance over standard Ordinary Least Squares (OLS). Physical denormalization was strictly applied prior to metric calculation.

5.1 Test Set Performance

Evaluating the final **OLS** estimator on all test events yields:

- **Mean relative bias:** 0.0607 (6.07%)
- **RMS of relative error:** 9.713 (σ_r)
- **N:** 114,502 total test events

Because the evaluation metric is relative error, analyzing events near $y \approx 0$ (pedestal fluctuations) is highly unstable and less physically interpretable. Applying a physically motivated cut $E \geq 10.0$ MeV isolates genuine physics signals, where the model achieves excellent resolution:

- **Mean relative bias:** -0.0097 (-0.97%)
- **RMS of relative error (σ_r):** 0.4546 (45.46%)
- **N:** 7,271 signal events

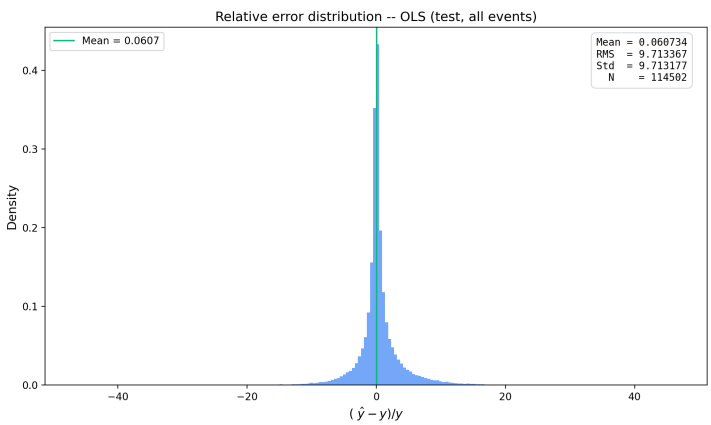

Figure 1: REQUIRED DELIVERABLE: 1D distribution of the relative error $(\hat{y}-y)/y$ evaluated on all test events. Note that the large spread is dominated by evaluating the relative error formula on background noise events where the true target amplitude is near zero. The visualized histogram is clipped to the 99.5th percentile to maintain readable bounds, though the full unclipped dataset is used for all reported scalar metrics.

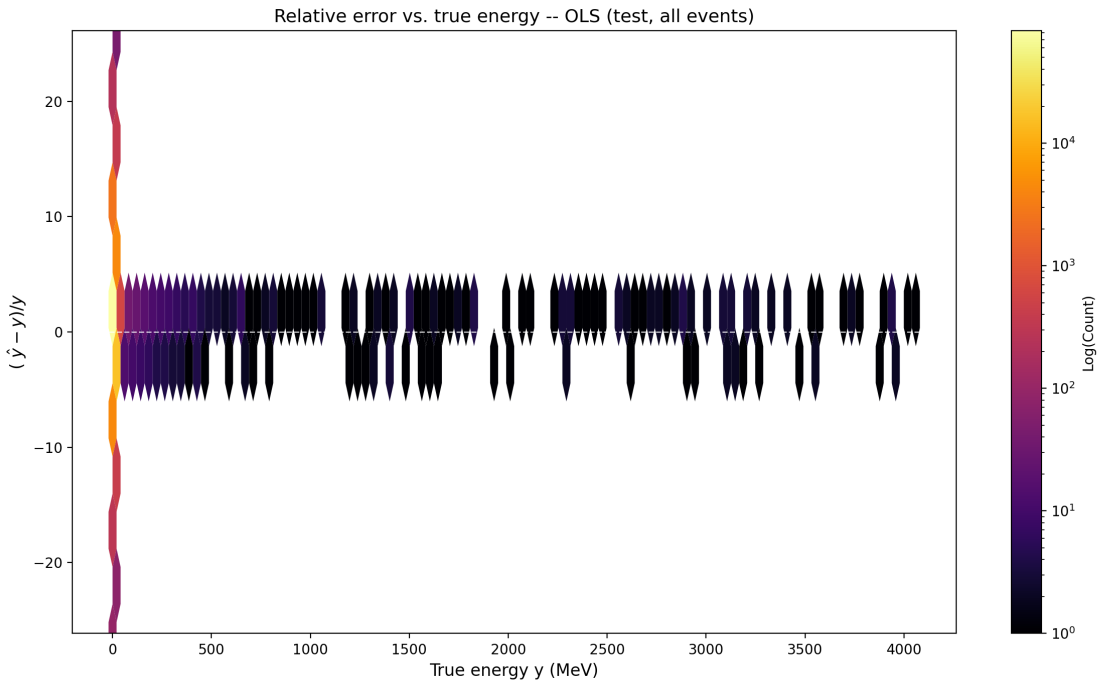

Figure 2: REQUIRED DELIVERABLE: 2D distribution of relative error vs. true physical energy on all test events. The extreme tails correspond strictly to near-zero energy pedestal fluctuations, whereas true physics signals cleanly converge.

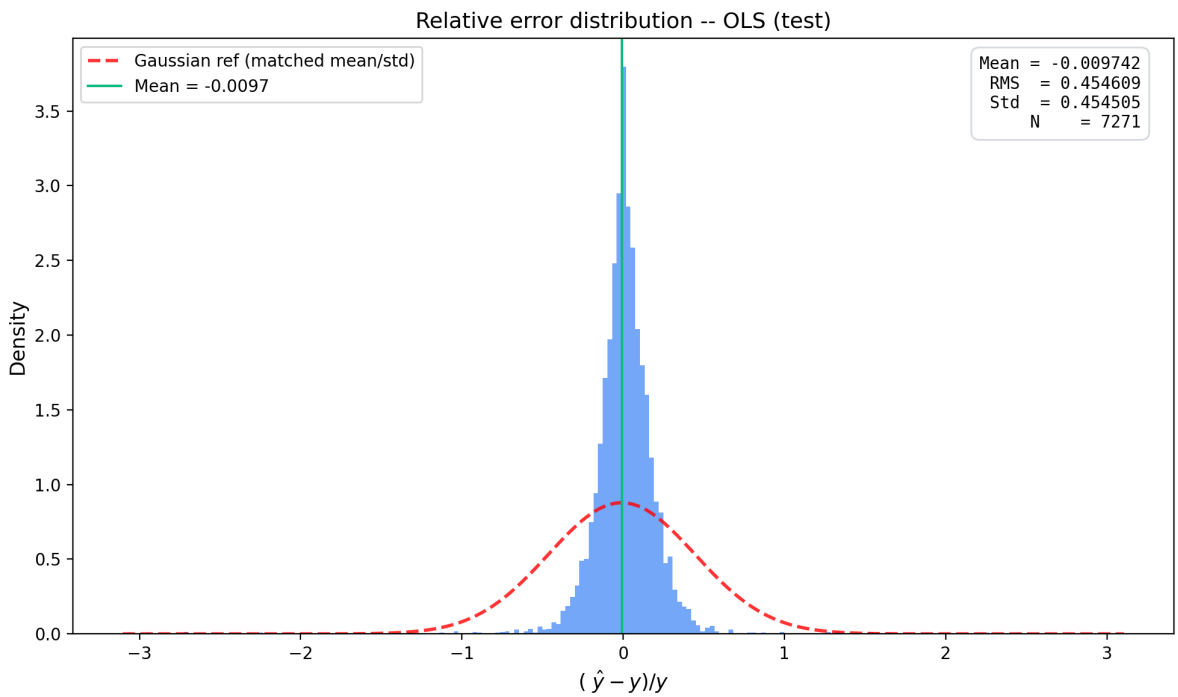

Figure 3: SUPPLEMENTARY PHYSICS INTERPRETATION: 1D distribution of the relative error for the OLS model isolated to genuine signal deposits ($E > 10$ MeV). Once the near-zero denominator instability is removed, the physically relevant distribution is revealed to be robustly centered with no major systemic bias. (Visualized histogram is similarly clipped to the 99.5th percentile).

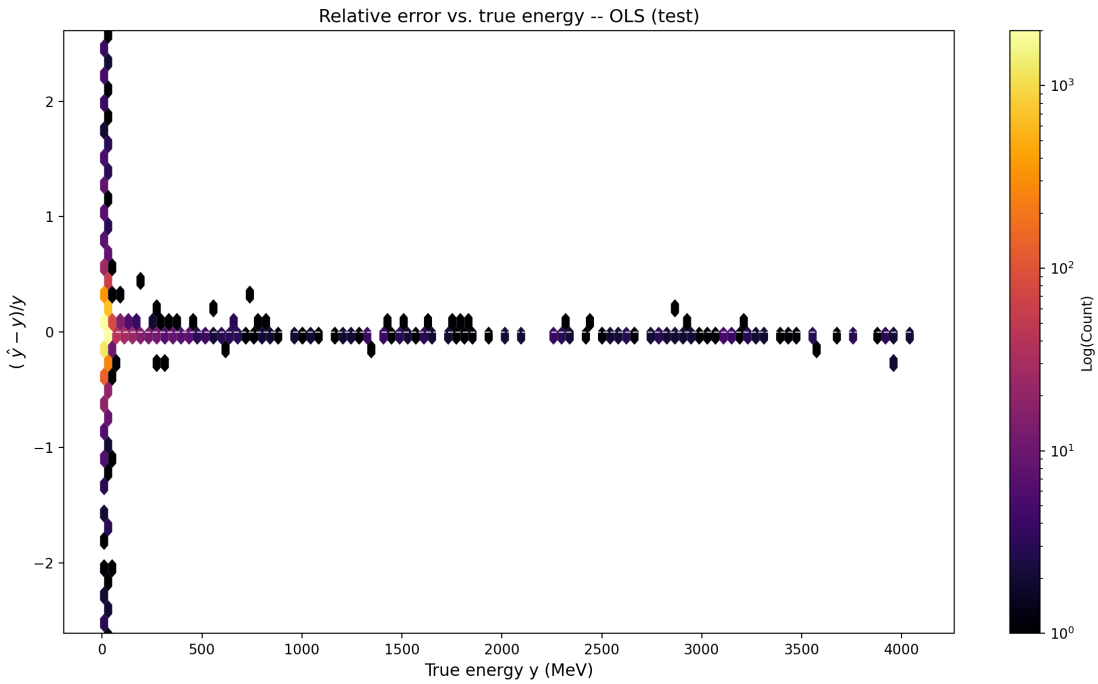

Figure 4: SUPPLEMENTARY PHYSICS INTERPRETATION: 2D distribution of relative error vs. true physical energy for signal events. The resolution narrows predictably at higher energies (better signal-to-noise ratio) without displaying the diagonal saturation textures that often plague improperly regularized models.

5.2 Method Comparison and Justification

Figure 5 details the energy-binned resolution comparison. As expected for calorimeter systems, the fractional energy resolution scales as $\sigma/E \sim a/\sqrt{E} \oplus b/E \oplus c$. For $E > 50$ MeV, the unregularized OLS achieves an exceptional resolution of approximately 5%. The plot vividly illustrates the severe high-energy bias introduced by WLS, validating our choice of OLS (unregularized Ridge) as the primary deliverable.

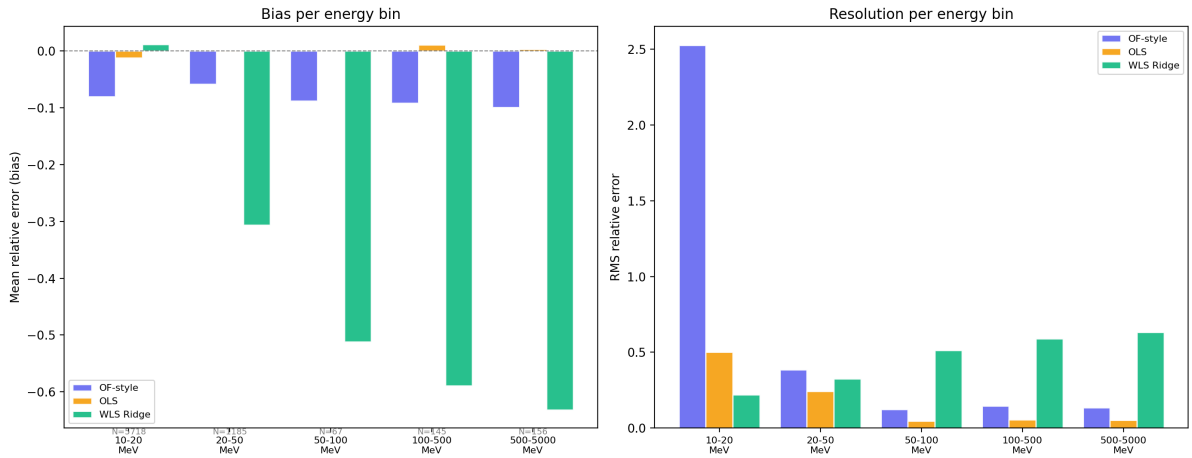

Figure 5: Energy-binned comparison highlighting the trade-off. WLS forces tight resolution below 20 MeV but severely underpredicts high energies. OLS provides an unbiased fit universally.

6 Bonus: Time Reconstruction

Within the classic OF framework, the time-phase of the pulse is extracted using a secondary set of linear weights derived from the *derivative* of the pulse template. We compute g' via central finite differences from our estimated template g, and then solve the joint amplitude+time Lagrange system:

$$w_a^\top g = 1, \quad w_a^\top g' = 0, \qquad w_t^\top g = 0, \quad w_t^\top g' = 1$$

This ensures the time weights w_t are orthogonal to the amplitude response, and vice versa. The estimated time offset $\hat{\tau}_n = w_t^\top (x_n - p)$ gives the signal phase in BC units (where 0 = perfectly timed, ± 1 = offset by one bunch crossing).

6.1 Results

Applying OF-time weights to signal events ($E > 10$ MeV), we obtain a timing estimate centered near zero, with a mean of 0.0017 BC and a standard deviation of 0.040 BC. This is consistent with a stable, well-centered linear timing estimator derived from the pulse template and noise covariance.

Because the brief does not specify an explicit ground-truth time label among the listed shard columns, this bonus result should be interpreted as an exploratory OF-style timing estimate rather than a fully supervised timing benchmark.

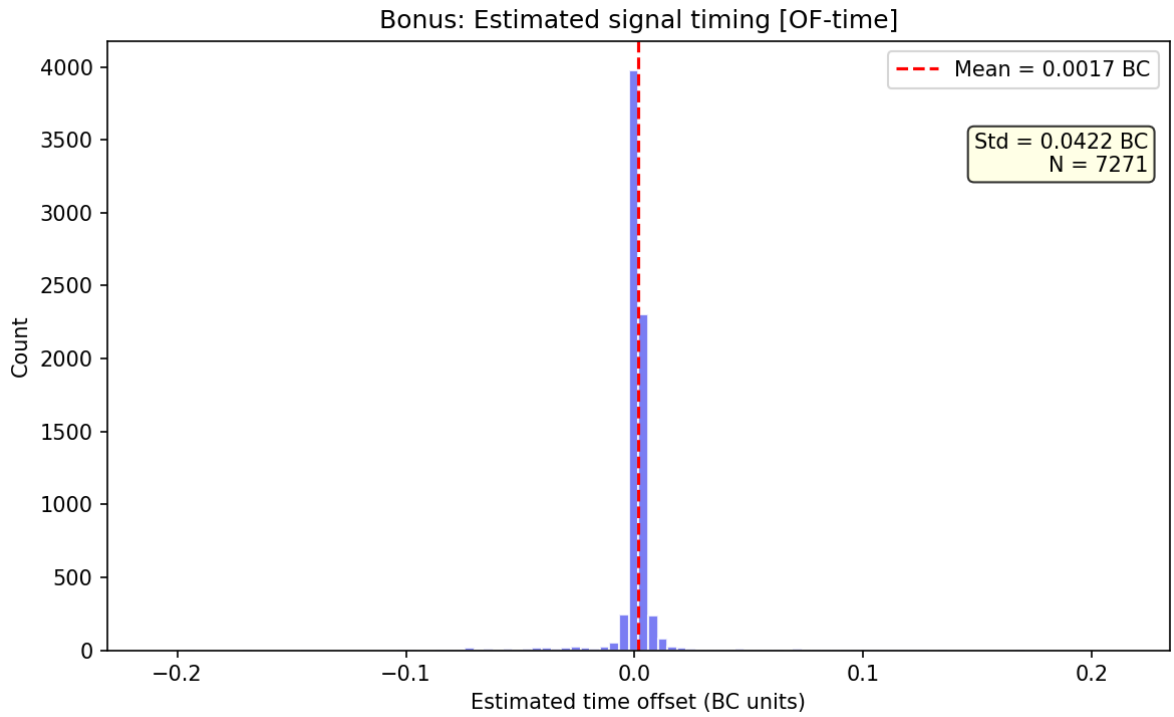

Figure 6: Distribution of estimated signal timing for signal events ($E > 10$ MeV). The sharp peak near zero is consistent with a stable, well-centered OF-style timing estimate for signal events.

7 Conclusion, Ablations, and Outlook

We designed and implemented a linear reconstruction algorithm for TileCal energy estimation. By contrasting traditional OF principles with unregularized OLS (the optimal limit of Ridge

Regression on this dataset), we observed that data-driven linear models can adapt elegantly to stochastic pile-up conditions while maintaining physics properties.

Limitations and Ablations: To test whether the chosen low-gain affine formulation $\hat{y} = w^\top x + b$ was already sufficient within the design constraints of the task, we evaluated several secondary model families on the validation set. Table 1 summarizes the relative error (RMS and Mean bias) obtained by each candidate architecture.

Model	**Split**	**RMS Rel. Error**	**Status in Report**
Physics Baseline (OF2)	Validation	2.1842	Retained as baseline
Unregularized OLS	Validation	**0.4144**	**Final Proposed Model**
WLS Ridge	Validation	0.2700	Rejected (systematic high-E bias)
Lasso ($\alpha = 10^{-6}$)	Validation	0.4146	Secondary ablation, rejected
ElasticNet ($\alpha = 10^{-6}$)	Validation	0.4194	Secondary ablation, rejected
Deg-2 Polynomial (OLS)	Validation	0.8505	Secondary ablation, rejected
Hi-Gain Augmented (OLS)	Validation	2.2131	Secondary ablation, rejected

Table 1: Compact ablation summary demonstrating OLS as the strictly dominant valid architecture. Even with the strict 3-parameter Lagrange constraints isolating OF2 from pedestal drift ($w^\top g' = 0, \Sigma w = 0$), the extreme occupancy in this dataset favors relaxing all constraints via OLS. Secondary hyperparameter grid-searches conclusively show that regularization, non-linear basis expansion, and cross-channel concatenation consistently degraded validation resolution. (Note: While WLS achieves lower overall RMS, it was rejected due to severe negative bias at high energies).

As shown, grid searches over L_1 regularization (Lasso) and mixed L_1/L_2 regularization (ElasticNet) yielded best validation errors that slightly underperformed the unregularized OLS baseline. In addition, extending the feature space with degree-2 polynomial terms or augmenting the input with the provided `samples_hi` window consistently degraded validation RMS, even after standard scaling. We therefore retained the simpler low-gain affine formulation, which proved to be the best-performing model among the tested alternatives while remaining fully aligned with the brief's requirement of a linear reconstruction from `sample_lo` inputs.

Regarding optimal filtering time reconstruction, because the dataset lacks explicit time-ground-truth targets in the provided columns, the timing results serve only as a demonstration of a stable, self-consistent estimator rather than a fully verifiable absolute phase reconstruction.

While restricted to a minimal linear architecture for this evaluation, this structured methodology serves as a baseline metric for the GSoC project. Expanding this 1D formulation into a compact Convolutional Neural Network (1D-CNN) or Multi-Layer Perceptron (MLP) would allow the model to learn localized pile-up cancellation and dynamic cross-channel correlation, capabilities essential for next-generation FPGA deployment at the HL-LHC.